Supplemental Material for: "Superfluid hydrodynamics from correlated many-body dynamics: Bridging variational Monte Carlo and Gross–Pitaevskii theory for microwave-shielded Bose gases"

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MODEL AND INTERACTION POTENTIAL

We consider N identical bosons of mass m in a cylindrically symmetric harmonic trap with Hamiltonian

$$\hat{H}(t) = \sum_{i=1}^{N}\left[-\frac{\hbar^2}{2m}\nabla_i^2 + V_{\rm tr}(\mathbf{r}_i, t)\right] + \sum_{i<j} v(\mathbf{r}_{ij}), \tag{S1}$$

where the trap potential is

$$V_{\rm tr}(\mathbf{r}, t) = \frac{1}{2}m\left[\omega_\rho(t)^2\rho^2 + \omega_z^2 z^2\right], \tag{S2}$$

with $\rho^2 = x^2 + y^2$ the cylindrical radial coordinate. We work in natural units $\hbar = m = 1$ and measure lengths in units of the oscillator length $a_{\rm ho} = \sqrt{\hbar/(m\omega_0)}$ with $\omega_0 = 1$.

The two-body interaction models microwave-shielded dipolar molecules [1]:

$$v(\mathbf{r}) = V_{\rm dip}(r, \theta) + V_{\rm shield}(r, \theta), \tag{S3}$$

where $r = |\mathbf{r}|$ and θ is the polar angle with respect to the z-axis. The dipolar term is

$$V_{\rm dip} = \frac{C_3}{r^3}(3\cos^2\theta - 1), \tag{S4}$$

and the repulsive shielding term is

$$V_{\rm shield} = \frac{C_6}{r^6}(\sin^4\theta + w_{12}\sin^2\theta\cos^2\theta), \tag{S5}$$

with $C_3 = 10^{-3}$, $C_6 = 10^{-6}$, and $w_{12} = 2.0$ in oscillator units. Both potentials are softened at short range by $r \to \sqrt{r^2 + \epsilon_r^2}$ with $\epsilon_r = 10^{-3}$ to avoid numerical singularities.

The system is initialized in the ground state of an isotropic trap $\omega_\rho = \omega_z = 1.0$. At $t = 0$, the radial trap frequency is instantaneously quenched to $\omega_\rho = 2.0$ while ω_z remains unchanged, exciting a radial breathing mode.

JASTROW–FEENBERG WAVEFUNCTION: FULL SPECIFICATION

One-body field u_1

The one-body field is parametrized with a separable radial-axial structure:

$$u_1(\rho, z, t) = u_\rho(\rho, t) + u_z(z, t), \tag{S6}$$

where each component consists of a harmonic reference plus Gaussian radial basis function corrections:

$$u_\rho(\rho, t) = -\frac{1}{2}(\omega_{\rm ref} + a_\rho(t))\rho^2 + \sum_{k=1}^{n_\rho} c_{\rho,k}(t)\, g_0(\rho; \ell_{\rho,k}), \tag{S7}$$

$$u_z(z, t) = -\frac{1}{2}(\omega_{\rm ref} + a_z(t))z^2 + \sum_{k=1}^{n_z} c_{z,k}(t)\, g_0(z; \ell_{z,k}), \tag{S8}$$

with $\omega_{\rm ref} = 1.0$ and the Gaussian basis functions

$$g_0(r;\ell) = \exp\left(-\frac{r^2}{2\ell^2}\right). \tag{S9}$$

The complex variational parameters $\{a_\rho(t), a_z(t), c_{\rho,1}(t), \ldots, c_{\rho,n_\rho}(t), c_{z,1}(t), \ldots, c_{z,n_z}(t)\}$ are time-dependent. Their real parts control the density profile and their imaginary parts encode the velocity field.

The length scales are distributed logarithmically:

$$\ell_k = \ell_{\rm min}\left(\frac{\ell_{\rm max}}{\ell_{\rm min}}\right)^{(k-1)/(n-1)}, \quad k = 1, \ldots, n, \tag{S10}$$

with $\ell_{\rm min} = 0.1$, $\ell_{\rm max} = 3.0$, and $n_\rho = n_z = 6$. This gives a total of $n_{u_1} = 2 + n_\rho + n_z = 14$ complex one-body parameters.

Two-body Jastrow u_2

The two-body factor is expanded in partial waves up to $\ell = 2$:

$$u_2(r, \cos\theta) = f_0(r) + f_2(r)\, P_2(\cos\theta), \tag{S11}$$

where $P_2(x) = (3x^2 - 1)/2$ is the Legendre polynomial. The radial components are

$$f_0(r) = u_{\rm cusp}(r) + \sum_{k=1}^{n_2} c_{0,k}\, g_0(r;\ell_{2,k}), \tag{S12}$$

$$f_2(r) = \sum_{k=1}^{n_2} c_{2,k}\, g_2(r;\ell_{2,k}), \tag{S13}$$

with $g_2(r;\ell) = r^2 \exp(-r^2/(2\ell^2))$ a regular $\ell = 2$ basis function and $n_2 = 10$ basis functions with length scales $\ell_2 \in [0.01, 0.7]$.

Cusp treatment

The short-range cusp function captures the two-body scattering boundary condition imposed by the C_6/r^6 repulsion:

$$u_{\rm cusp}(r) = -\frac{\kappa}{\sqrt{r^2 + \epsilon_c^2}}\, \exp\left(-\frac{r^2}{2r_c^2}\right), \tag{S14}$$

with cusp strength $\kappa = 0.1$, softening parameter $\epsilon_c = 5 \times 10^{-3}$, and cutoff radius $r_c = 0.01$. The $1/r$ singularity is regularized by ϵ_c to avoid numerical overflow, while the Gaussian envelope ensures that the cusp affects only the short-range pair structure ($r \lesssim r_c$). These parameters are fixed (not variational) and determined by the scattering properties of the two-body potential.

The total variational space has $n_{u_1} + 2n_2 = 14 + 20 = 34$ parameters, of which $n_{u_1} = 14$ one-body parameters are time-dependent and $2n_2 = 20$ two-body parameters are fixed in the frozen-u_2 approximation.

TIME-DEPENDENT VARIATIONAL PRINCIPLE: DETAILED DERIVATION

General TDVP formulation

The TDVP projects the Schrödinger equation onto the variational tangent space spanned by derivatives of $\ln\Psi$ with respect to the variational parameters $\{\theta_\alpha\}$. Starting from the Schrödinger equation $i\partial_t\Psi = \hat{H}\Psi$ and the variational ansatz $\Psi(\mathbf{X};\boldsymbol{\theta}(t))$, the McLachlan variational principle [2] minimizes

$$\delta\left\| i\partial_t\Psi - \hat{H}\Psi \right\|^2 = 0 \tag{S15}$$

with respect to $\dot{\boldsymbol{\theta}}$. This yields the equations of motion [3]:

$$\sum_{\beta} S_{\alpha\beta}\,\dot{\theta}_{\beta} = -i\,F_{\alpha}, \tag{S16}$$

where the overlap matrix (quantum geometric tensor) and force vector are

$$S_{\alpha\beta} = \langle \Delta O_{\alpha}^{*}\,\Delta O_{\beta}\rangle_t = \langle O_{\alpha}^{*} O_{\beta}\rangle_t - \langle O_{\alpha}^{*}\rangle_t \langle O_{\beta}\rangle_t, \tag{S17}$$

$$F_{\alpha} = \langle \Delta O_{\alpha}^{*}\,\Delta E_L\rangle_t = \langle O_{\alpha}^{*} E_L\rangle_t - \langle O_{\alpha}^{*}\rangle_t \langle E_L\rangle_t, \tag{S18}$$

with logarithmic derivatives $O_{\alpha} = \partial_{\theta_\alpha} \ln \Psi$.

Local energy

For the Jastrow ansatz $\Psi = e^{U}$ with $U = \sum_i u_1(\mathbf{r}_i, t) + \sum_{i<j} u_2(r_{ij})$, the local energy is

$$E_L = \frac{\hat{H}\Psi}{\Psi} = \sum_{i=1}^{N}\left[-\frac{1}{2}(\nabla_i^2 U + |\nabla_i U|^2) + V_{\mathrm{tr}}(\mathbf{r}_i, t)\right] + \sum_{i<j} v(\mathbf{r}_{ij}), \tag{S19}$$

where

$$\nabla_i U = \nabla u_1(\mathbf{r}_i, t) + \sum_{j\neq i} \nabla_i u_2(r_{ij}), \tag{S20}$$

$$\nabla_i^2 U = \nabla^2 u_1(\mathbf{r}_i, t) + \sum_{j\neq i} \nabla_i^2 u_2(r_{ij}). \tag{S21}$$

Monte Carlo estimation

The expectation values in Eqs. (S17)–(S18) are estimated by sampling configurations $\mathbf{X}^{(s)}$ from $|\Psi(\mathbf{X}, t)|^2$ using the Metropolis algorithm. We employ single-particle moves: for each walker and each particle i, a displacement $\Delta\mathbf{r} \sim \mathrm{Uniform}(-\delta/2, \delta/2)^3$ is proposed with step size $\delta = 0.45$, and accepted with probability $\min(1, |\Psi_{\mathrm{new}}|^2/|\Psi_{\mathrm{old}}|^2)$.

We use $N_w = 65{,}536$ parallel walkers (GPU-accelerated using JAX [4]), with $n_{\mathrm{burn}} = 30$ equilibration sweeps followed by $n_{\mathrm{meas}} = 10$ measurement sweeps separated by 3 thinning sweeps, per RK4 stage.

The TDVP linear system (S16) is solved via eigendecomposition of S:

$$\dot{\theta}_{\alpha} = -i \sum_{\beta} S_{\alpha\beta}^{-1} F_{\beta}, \tag{S22}$$

with eigenvalue truncation at $\lambda_{\min}/\lambda_{\max} = 10^{-4}$ and a diagonal shift $S \to S + \eta\mathbb{1}$ with $\eta = 10^{-2}$.

For imaginary-time propagation (ground-state preparation), we use $\dot{\theta} = -F/S$ with $d\tau = 10^{-3}$ for 2000 steps. For real-time evolution after the quench, the step size is $dt = 2 \times 10^{-4}$ with adaptive time-stepping that limits the RMS parameter change to $\|\dot{\theta}\,dt\|_{\mathrm{rms}} < 0.01$.

FROZEN-u_2 APPROXIMATION: DETAILED DERIVATION OF HYDRODYNAMIC EQUATIONS

Motivation: stiffness of u_2

Our tVMC simulations reveal that during the trap quench $\omega_\rho : 1 \to 2$, the two-body Jastrow parameters $(c_{0,k}, c_{2,k})$ change by less than 1% of their ground-state values, while the one-body parameters $(a_\rho, a_z, c_{\rho,k}, c_{z,k})$ undergo order-unity changes. This reflects the physical expectation that short-range pair correlations equilibrate much faster than the global density redistribution driven by the trap quench.

TDVP with continuous one-body field

Freezing u_2 at its ground-state value (real, time-independent) and promoting $u_1(\mathbf{r},t)$ to a continuous variational field, the logarithmic derivative becomes the microscopic density operator:

$$O(\mathbf{r};\mathbf{X}) = \frac{\delta \ln \Psi}{\delta u_1(\mathbf{r},t)} = \sum_{i=1}^{N} \delta^{(3)}(\mathbf{r}-\mathbf{r}_i) \equiv \hat{\rho}(\mathbf{r}). \tag{S23}$$

The TDVP equation (S16) becomes a continuum integral equation:

$$\int d^3r'\; S(\mathbf{r},\mathbf{r}';t)\, \partial_t u_1(\mathbf{r}',t) = -i\, F(\mathbf{r};t), \tag{S24}$$

where

$$S(\mathbf{r},\mathbf{r}';t) = \langle \Delta\hat{\rho}(\mathbf{r})\, \Delta\hat{\rho}(\mathbf{r}')\rangle_t, \tag{S25}$$

$$F(\mathbf{r};t) = \langle \Delta\hat{\rho}(\mathbf{r})\, \Delta E_L\rangle_t. \tag{S26}$$

The density–density correlator can be decomposed as

$$S(\mathbf{r},\mathbf{r}') = \rho(\mathbf{r})\delta^{(3)}(\mathbf{r}-\mathbf{r}') + \rho^{(2)}(\mathbf{r},\mathbf{r}') - \rho(\mathbf{r})\rho(\mathbf{r}'), \tag{S27}$$

where $\rho(\mathbf{r}) = \langle \hat{\rho}(\mathbf{r})\rangle_t$ is the one-body density and $\rho^{(2)}(\mathbf{r},\mathbf{r}')$ is the two-body density.

Decomposition into amplitude and phase

We decompose the complex one-body field:

$$u_1(\mathbf{r},t) = a(\mathbf{r},t) + i\theta(\mathbf{r},t), \qquad a,\theta \in \mathbb{R}. \tag{S28}$$

Because u_2 is real and frozen, the many-body wavefunction has the structure

$$\Psi = e^{\sum_i a(\mathbf{r}_i) + i\sum_i \theta(\mathbf{r}_i) + \sum_{i<j} u_2(r_{ij})}, \tag{S29}$$

so the many-body phase is purely one-body: $\arg\Psi = \sum_i \theta(\mathbf{r}_i)$. The flow is therefore irrotational with velocity

$$\mathbf{v}(\mathbf{r},t) = \nabla\theta(\mathbf{r},t). \tag{S30}$$

No backflow terms arise because u_2 carries no time-dependent phase.

Internal energy functional

Define the internal energy as the expectation value of kinetic plus interaction energy in the frozen-u_2 state at density ρ:

$$E_{\text{int}}[\rho;u_2] = \langle \hat{T} + \hat{V}_{\text{int}}\rangle_t. \tag{S31}$$

If E_{int} depends on ρ only through the local density (LDA), then $E_{\text{int}}^{\text{LDA}} = \int \rho(\mathbf{r})\varepsilon(\rho(\mathbf{r}))\, d^3r$, where $\varepsilon(\rho)$ is the energy per particle of the uniform system. The TDVP action then takes the form

$$\mathcal{S} = \int dt\, d^3r \left[\rho\, \partial_t\theta - \frac{1}{2}\rho|\nabla\theta|^2 - \rho\, V_{\text{tr}} - \rho\, \varepsilon(\rho)\right], \tag{S32}$$

and the stationarity conditions $\delta\mathcal{S}/\delta\theta = 0$ and $\delta\mathcal{S}/\delta a = 0$ (where ρ depends on a) yield the closed hydrodynamic system:

$$\text{Continuity:} \quad \partial_t\rho + \nabla\cdot(\rho\mathbf{v}) = 0, \tag{S33}$$

$$\text{Euler:} \quad \partial_t\mathbf{v} + \nabla\left(\frac{1}{2}v^2 + V_{\text{tr}} + \mu(\rho)\right) = 0, \tag{S34}$$

where the chemical potential is

$$\mu(\rho) = \frac{d[\rho\varepsilon(\rho)]}{d\rho} = \varepsilon(\rho) + \rho\varepsilon'(\rho). \tag{S35}$$

Gross–Pitaevskii reformulation

Equations (S33)–(S34) can be rewritten as a single GPE for $\psi = \sqrt{\rho}\, e^{i\theta}$:

$$i\partial_t \psi = \left[-\frac{1}{2}\nabla^2 + V_{\rm tr} + \mu(|\psi|^2) \right] \psi, \tag{S36}$$

where the Laplacian provides both the quantum pressure ($Q = -\frac{1}{2}\nabla^2\sqrt{\rho}/\sqrt{\rho}$) and the flow kinetic energy. This formulation avoids the $\rho \to 0$ singularity of the Madelung variables and provides a single complex equation.

Quantum pressure

The quantum pressure term arises from the density dependence of the overlap kernel $S(\mathbf{r}, \mathbf{r}')$ beyond the LDA. In the GPE formulation (S36), it is automatically included through the $-\frac{1}{2}\nabla^2\psi$ term. For the weakly interacting $N = 8$ system ($E_{\rm int}/E_{\rm tot} \approx 3.5\%$), quantum pressure provides the dominant repulsive mechanism. Without it, the dynamics is qualitatively incorrect. For $N = 128$ ($E_{\rm int}/E_{\rm tot} \approx 21\%$), both quantum pressure and the EOS contribute significantly.

EQUATION OF STATE: VMC PROCEDURE AND FITTING

Uniform-system VMC

To extract the EOS, we perform VMC calculations for a uniform system of N particles in a cubic box with periodic boundary conditions at a grid of densities. The two-body Jastrow $u_2(r, \hat{\mathbf{r}})$ is frozen at the ground-state values from the trapped-system tVMC, while u_1 is taken as constant.

At each density, we use $N_w = 8{,}192$ walkers with 200 equilibration sweeps and 500 measurement sweeps (thinning by 5) to estimate the energy per particle $\varepsilon(\rho)$. Error bars are obtained from a blocking analysis.

For $N = 8$, we scan 20 densities logarithmically over $\rho \in [0.05, 3.0]$. For $N = 128$, the higher peak densities encountered during compression require extending the range to $\rho \in [0.05, 15.0]$ with 30 density points.

Polynomial fitting

We fit $\varepsilon(\rho)$ to a polynomial using weighted least-squares:

$$\varepsilon(\rho) = a_1\rho + a_2\rho^2 + a_3\rho^3. \tag{S37}$$

Model selection between quadratic ($a_3 = 0$) and cubic is performed using the Akaike information criterion (AIC).

For $N = 8$, the cubic fit is preferred:

$$a_1 = 0.102, \quad a_2 = 0.023, \quad a_3 = -0.007, \tag{S38}$$

with reduced $\chi^2_\nu = 0.41$. The negative a_3 introduces a stability limit $d\mu/d\rho = 0$ at $\rho_{\rm stab} \approx 2.67$, above which we regularize by linear extrapolation.

For $N = 128$, a quadratic suffices:

$$a_1 = 0.111, \quad a_2 = 0.001, \quad a_3 = 0, \tag{S39}$$

which is unconditionally stable ($d\mu/d\rho > 0$ for all ρ).

Emergent effective interactions

The chemical potential $\mu(\rho) = 2a_1\rho + 3a_2\rho^2 + \cdots$ admits a physically transparent interpretation. Comparing with the standard GPE nonlinearity $\mu = g\rho$:

- The leading term $2a_1\rho$ defines a *renormalized two-body coupling* $g_{\rm eff} = 2a_1$ that absorbs the effect of frozen pair correlations into an effective contact interaction.
- The next-order term $3a_2\rho^2$ has the structure of an *effective three-body interaction* $g_3|\psi|^4$ with $g_3 = 3a_2$.

This three-body term is not a genuine microscopic three-body force but emerges from integrating out the density-dependent pair correlations encoded in u_2. The resulting GPE

$$i\partial_t\psi = \left[-\frac{1}{2}\nabla^2 + V_{\rm tr} + g_{\rm eff}|\psi|^2 + g_3|\psi|^4\right]\psi \tag{S40}$$

provides a concrete, non-perturbative realization of the extended GPE discussed in the context of quantum droplets [5] and dipolar condensates [6].

HYDRODYNAMIC PDE SOLVER: NUMERICAL DETAILS

Cylindrical coordinates

Exploiting azimuthal symmetry, the 3D GPE reduces to 2D in cylindrical coordinates (r, z):

$$\nabla^2\psi = \frac{\partial^2\psi}{\partial r^2} + \frac{1}{r}\frac{\partial\psi}{\partial r} + \frac{\partial^2\psi}{\partial z^2}. \tag{S41}$$

For $N = 8$: domain $r \in [0, 5]$, $z \in [-5, 5]$, grid 128×256.
For $N = 128$: domain $r \in [0, 6]$, $z \in [-8, 8]$, grid 192×512.

Boundary conditions

At $r = 0$: azimuthal symmetry requires $\partial_r\psi|_{r=0} = 0$, implemented via a ghost point $\psi(-\Delta r) = \psi(\Delta r)$. The $(1/r)\partial_r\psi$ term is evaluated using L'Hôpital's rule: $\lim_{r\to 0}(1/r)\partial_r\psi = \partial_r^2\psi|_{r=0}$.
At outer boundaries ($r = R_{\rm max}$, $z = \pm Z_{\rm max}$): Dirichlet $\psi = 0$.

Fourth-order finite differences

Interior derivatives use fourth-order central differences:

$$f_i'' = \frac{-f_{i-2} + 16f_{i-1} - 30f_i + 16f_{i+1} - f_{i+2}}{12\Delta x^2}, \tag{S42}$$

$$f_i' = \frac{-f_{i+2} + 8f_{i+1} - 8f_{i-1} + f_{i-2}}{12\Delta x}. \tag{S43}$$

Near boundaries, ghost-point values are determined by symmetry or Dirichlet conditions. The fourth-order scheme is critical for frequency accuracy: the second-order scheme gives $\omega_{\rm br} = 3.99$, while fourth-order gives 4.19 (matching tVMC).

Time integration

RK4 with dispersive CFL condition:

$$\Delta t \le C_{\rm CFL}\frac{(\min\Delta x)^2}{\pi}, \quad C_{\rm CFL} = 0.3. \tag{S44}$$

An additional constraint $\Delta t \le 2C_{\rm CFL}/|\mu_{\rm max}|$ limits phase accumulation per step.

Imaginary-time ground state

The GPE ground state is obtained by imaginary-time propagation:

$$\psi \to \psi - d\tau\, \hat{H}_{\rm GPE}\psi, \qquad \psi \to \psi\sqrt{N/\|\psi\|^2}, \tag{S45}$$

with $d\tau = 0.3\Delta x^2/\pi$, iterated until $\delta\mu < 10^{-8}$. Starting from a Gaussian guess, convergence is achieved in $\sim$ 30,000 iterations for $N = 8$ and $\sim$ 45,000 for $N = 128$.

Observables

At regular intervals we compute (using cylindrical trapezoidal integration $\int f\, 2\pi r\, dr\, dz$):

$$N(t) = \int |\psi|^2\, 2\pi r\, dr\, dz, \tag{S46}$$

$$\langle\rho^2\rangle(t) = \frac{1}{N}\int r^2|\psi|^2\, 2\pi r\, dr\, dz, \tag{S47}$$

$$\langle z^2\rangle(t) = \frac{1}{N}\int z^2|\psi|^2\, 2\pi r\, dr\, dz, \tag{S48}$$

$$E(t) = \int\left[-\frac{1}{2}\psi^*\nabla^2\psi + V|\psi|^2 + |\psi|^2\varepsilon(|\psi|^2)\right] 2\pi r\, dr\, dz. \tag{S49}$$

The kinetic energy uses the same discrete Laplacian as the dynamics, ensuring consistent energy conservation.

QUANTITATIVE COMPARISON: TVMC VS GPE

Table S1 summarizes the comparison between frozen-u_2 tVMC and the GPE solver for both $N = 8$ and $N = 128$ after the quench $\omega_\rho : 1 \to 2$. For tVMC, we report the energy conservation as the drift in the stochastic energy estimator after the quench; the instantaneous fluctuation is $\sim$ 5% for $N = 8$ and $\sim$ 7% for $N = 128$ due to Monte Carlo noise, but the systematic drift is negligible.

TABLE S1. Comparison of frozen-u_2 tVMC and GPE (imaginary-time ground state, fourth-order Laplacian) for the quench $\omega_\rho : 1 \to 2$. $\delta E/E$ for tVMC is the energy drift (not stochastic fluctuation); for GPE it is the deterministic conservation error.

	$N = 8$		$N = 128$	
	tVMC	GPE	tVMC	GPE
$E_{\rm gs}$	12.42	12.44	276.8	269.0
$\langle\rho^2\rangle_0$	1.051	1.058	1.555	1.561
$\langle z^2\rangle_0$	0.525	0.529	0.772	0.780
$\omega_{\rm br}$	4.19	4.19	4.19	4.19
$\langle\rho^2\rangle$ range	$[0.26, 1.06]$	$[0.26, 1.06]$	$[0.37, 1.56]$	$[0.37, 1.56]$
$\langle z^2\rangle$ range	$[0.52, 0.61]$	$[0.53, 0.61]$	$[0.59, 2.05]$	$[0.56, 2.00]$
$\lvert\delta E/E\rvert$	$< 0.01\%$	0.1%	2.8%	0.05%
$\max\lvert\delta N/N\rvert$	—	7×10^{-4}	—	3×10^{-4}

CONVERGENCE TESTS

Spatial convergence

Table S2 shows conservation diagnostics for $N = 8$ at different grid resolutions and Laplacian orders.

TABLE S2. Conservation diagnostics for the GPE simulation ($N = 8$, $t \in [0, 3]$).

Grid	∇^2 order	$\max\lvert\delta N/N\rvert$	$\max\lvert\delta E/E\rvert$
128×256	2nd	1.09×10^{-3}	1.20×10^{-3}
128×256	4th	7.30×10^{-4}	1.29×10^{-3}
160×320	4th	4.65×10^{-4}	8.22×10^{-4}

Temporal convergence

Halving the CFL factor from 0.3 to 0.15 (doubling the time steps) produces identical conservation to machine precision, confirming that the RK4 temporal error is negligible and the conservation errors are purely spatial.

Conservation behavior

The conservation errors are *oscillatory*: they peak during maximum compression (steepest density gradients) and nearly vanish when the cloud re-expands. At $t = 1.5$ (full expansion for $N = 8$), $|\delta N/N| = 3.2 \times 10^{-6}$ and $|\delta E/E| = 2.8 \times 10^{-6}$.

For $N = 128$, the larger grid (192×512) achieves $\max|\delta N/N| = 3.1 \times 10^{-4}$ and $\max|\delta E/E| = 4.5 \times 10^{-4}$ over $t \in [0, 6]$.

SCALING ANSATZ

As a further benchmark, we compare with the Castin–Dum scaling ansatz [7, 8]. Assuming self-similar evolution $\rho(\mathbf{r}, t) = (\lambda_r^2 \lambda_z)^{-1} \rho_0(r/\lambda_r, z/\lambda_z)$, the scaling parameters satisfy

$$\ddot{\lambda}_r = -\omega_\rho^2 \lambda_r + \frac{\omega_{\rho,0}^2}{\lambda_r^3 \lambda_z}, \tag{S50}$$

$$\ddot{\lambda}_z = -\omega_z^2 \lambda_z + \frac{\omega_{z,0}^2}{\lambda_r^2 \lambda_z^2}. \tag{S51}$$

This is the non-interacting ($\gamma = 1$) limit, appropriate for benchmarking the trap-dominated regime. The scaling ansatz predicts a breathing frequency close to $2\omega_\rho$ but overestimates the oscillation amplitude because it neglects quantum pressure corrections to the self-similar assumption.

COMPARISON BETWEEN FROZEN-u_2 TVMC AND FULL TVMC

For $N = 8$, we verify that the frozen-u_2 approximation is accurate by comparing with a full tVMC simulation where all 34 parameters evolve. Table S3 summarizes the comparison.

TABLE S3. Comparison of full and frozen-u_2 tVMC for $N = 8$.

	Full tVMC	Frozen-u_2 tVMC
$E_{\rm gs}$	12.433	12.423
$\langle\rho^2\rangle_0$	1.049	1.051
$\langle z^2\rangle_0$	0.525	0.525
$\langle\rho^2\rangle$ range	$[0.264, 1.060]$	$[0.264, 1.055]$
$\langle z^2\rangle$ range	$[0.522, 0.613]$	$[0.522, 0.613]$
$\omega_{\rm br}$ (dominant)	3.14	4.19

The ground state and oscillation amplitudes agree to within 0.5%. The breathing frequency differs: the full tVMC shows a dominant mode at $\omega \approx 3.14$ due to additional mode coupling through the time-dependent u_2, while the

frozen-u_2 tVMC and GPE both give $\omega \approx 4.19 \approx 2\omega_\rho$, consistent with the hydrodynamic prediction for this weakly interacting regime.

For $N = 128$, only the full tVMC simulation was performed (the frozen-u_2 tVMC was not run separately). The GPE with the VMC-derived EOS reproduces the full tVMC breathing frequency ($\omega = 4.19$) and oscillation amplitudes to within 2%, as discussed in the main text.

PARTIAL-WAVE CONVERGENCE: $\ell = 4$ ANALYSIS

The microwave-shielded interaction [Eq. (2) of the main text] contains two angular structures: the dipolar term $\propto 3\cos^2\theta - 1 = 2P_2(\cos\theta)$ and the shielding term $\propto \sin^4\theta$. While the dipolar term couples only $\ell = 0$ and $\ell = 2$, the $\sin^4\theta$ term contains a P_4 component and in principle requires $\ell = 4$ in the Jastrow expansion:

$$u_2(r, \cos\theta) = f_0(r) + f_2(r)\, P_2(\cos\theta) + f_4(r)\, P_4(\cos\theta). \tag{S52}$$

To test whether $\ell = 4$ matters, we performed a ground-state optimization for $N = 8$ with $n_4 = 6$ additional $\ell = 4$ basis functions $g_4(r; \ell) = r^4 \exp(-r^2/(2\ell^2))$. Table S4 compares the ground-state properties.

TABLE S4. Ground-state comparison with and without $\ell = 4$ ($N = 8$).

	$\ell \leq 2$ (v7)	$\ell \leq 4$ (v8)
$E_{\rm gs}$	12.423	12.439
$\langle\rho^2\rangle_0$	1.051	1.051
$\langle z^2\rangle_0$	0.525	0.525
$\lvert c_0\rvert_{\rm rms}$	1.503	1.503
$\lvert c_2\rvert_{\rm rms}$	0.046	0.046
$\lvert c_4\rvert_{\rm rms}$	—	0.001
$\lvert c_4/c_0\rvert$	—	0.07%

The $\ell = 4$ contribution is negligible: $|c_4|_{\rm rms}$ is 0.07% of $|c_0|_{\rm rms}$, and even the $\ell = 2$ anisotropy is only 3% of $\ell = 0$. The ground-state energy and moments are unaffected to within the stochastic error of the VMC.

Dynamical stability of $\ell = 4$ parameters

During real-time quench dynamics, the weakly constrained c_4 parameters are susceptible to numerical instability in the TDVP. With standard regularization ($\eta = 10^{-4}$), $|c_4/c_0|$ grows from 0.07% to $\sim$ 39% over $t \in [0, 3]$, despite negligible physical coupling. Two diagnostic tests confirm this is a numerical artifact:

1. *No-quench test*: With ω_ρ held at 1.0 (no perturbation), $|c_4/c_0|$ still grows from 0.02% to 0.48%—a monotonic random walk that should be identically zero for a stationary state.

2. *Regularization sensitivity*: Increasing the diagonal shift from $\eta = 10^{-4}$ to $\eta = 10^{-3}$ suppresses the growth to 1.4%, confirming noise-driven amplification of weakly constrained directions.

These tests justify our choice to retain only $\ell \leq 2$ in the production simulations. For the frozen-u_2 approximation used to derive the hydrodynamic equations, this instability is entirely avoided since the pair parameters are not evolved.

EOS FITTING: EXTENDED RANGE FOR $N = 128$

The $N = 128$ system reaches peak densities $\rho_{\rm peak} \approx 29$ during maximum radial compression. The EOS must therefore be reliable over a much wider density range than for $N = 8$.

We extended the uniform VMC scan to $\rho_{\rm max} = 15$ with 30 density points. The results are well described by a quadratic: $\varepsilon(\rho) = 0.111\rho + 0.001\rho^2$. The EOS is nearly linear, reflecting the dominance of the long-range dipolar interaction at high density.

We compared both quadratic (poly2) and cubic (poly3) fits. The poly3 fit has a stability limit at $\rho_{\text{stab}} \approx 10.7$, beyond which $d\mu/d\rho < 0$. Since the simulation reaches $\rho \approx 29$, this requires regularization. The poly2 fit is unconditionally stable and gives slightly better agreement with tVMC—particularly in the axial dynamics, where $\langle z^2 \rangle_{\max} = 1.94$ (poly2) vs. 1.84 (poly3) compared to tVMC 2.05.

The small quadratic coefficient ($a_2/a_1 \approx 0.007$) means the effective three-body coupling $g_3 = 3a_2 \approx 0.002$ is weak relative to the two-body coupling $g_{\text{eff}} = 2a_1 \approx 0.22$. The beyond-mean-field correction is therefore small but non-negligible, contributing $\sim 1\%$ of the chemical potential at typical densities.

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